



WEEK 2

# Company Valuation

How a stock's price is actually built, how to judge a business before you buy it, and how to manage what you own toward a goal.

---



# Round the table

1

Anything new you did with money this week — an account, a payment, a purchase decision?

2

Something interesting you learned about money or finance?

3

A new investment you're looking at or like right now?

**This week:** Bought or sold a stock lately — what was the actual reason: a number, a story, a price move, or a tip?



# Pick your path

|    |                                              |                 |       |
|----|----------------------------------------------|-----------------|-------|
| M1 | What actually moves a stock price            |                 | 6 min |
| M2 | Reading the business — statements & earnings |                 | 7 min |
| M3 | Valuation I — multiples (relative value)     |                 | 7 min |
| M4 | Valuation II — DCF & forecasting             |                 | 7 min |
| M5 | Quality & the buy checklist                  | most actionable | 6 min |
| M6 | How hedge funds & quants play it             | optional        | 5 min |
| M7 | Technical analysis & timing                  | optional        | 6 min |
| M8 | Building & managing the portfolio            |                 | 7 min |



# Two suggested routes

## Route A · Think like an investor

~30 min · lighter math

- M1 → M5 → M6 → M7 → M8
- What makes a great business, what funds actually do, how timing works, how to run a portfolio
- Concept & strategy

## Route B · Value a company with me

deep dive · runs long

- M1 → M2 → M3 → M4 → M8
- NVIDIA's Q1 FY2027 filing on screen — compute the statements, multiples & a DCF live
- Hands-on mechanics

Both share the same bookends — M1 (why prices move) and M8 (build & manage the portfolio) — and differ only in the middle.

M1

# What actually moves a stock price

Price vs. value, the total-return equation, expectations, and the discount-rate channel.

# Price vs. value

They're tied by a long, loose leash — not a rigid rod.

- Price is what someone will pay right now — set by mood, momentum, liquidity, and flows.
- Value is the present value of the cash the business will generate — growth, margins, risk.
- Mr. Market quotes you a buy/sell price every day — euphoric some days, despairing others.
  - His mood is a service to transact against, not a guide to what your stake is worth.
- “The market is wrong” only pays if you can name which input — growth, margin, rate, duration — you disagree with.

## TERMS ON THIS SLIDE

**PV** — Present Value  
today's worth of future cash

**Mr. Market** — Graham's manic-depressive partner who quotes a price daily

# The master equation — total return

$\text{TSR} \approx \text{dividend yield} + \text{EPS growth} \pm \Delta\text{P/E}$

# 16.6%

S&P 500 total return per  
year, 2012–2021

EPS +7.4%

P/E re-rating +6.9%

Div +2.3%

Nearly **half** that decade's return was **re-rating** — paying a higher multiple. Volatile and mean-reverting; it can't compound forever. EPS growth is the durable engine.

## TERMS ON THIS SLIDE

**TSR** — Total Shareholder Return  
 $\approx \text{div yield} + \text{EPS growth} \pm \Delta\text{P/E}$

**EPS** — Earnings Per Share  
 $= \text{net income} \div \text{diluted shares}$

**P/E** — Price-to-Earnings  
 $= \text{price} \div \text{EPS} \cdot \text{"re-rating"} = \Delta\text{P/E}$



# Why a beat can still tank a stock

**+85%** ▲ beat

NVDA revenue YoY — record \$81.6B; a beat on revenue, EPS and the outlook

Q1 FY2027

**~\$91B**

next-quarter guide ( $\pm 2\%$ ) — the number that actually mattered

**-2%** ▼ fell

the stock the next session — 4th straight beat-and-fall; shares had run ~20% into the print



# The discount-rate channel

A stock is worth the PV of future cash, discounted at  $r$  = risk-free rate + equity risk premium.

- Gordon shorthand:  $P/E \approx \text{payout} \div (r - g)$ .
  - More cash back → worth more; higher required return → pay less; faster growth → pay more.
- \$2 payout,  $r$  8%,  $g$  4% → \$50. Nudge  $r$  to 9% → \$40 — a 20% drop on the same business.
- Growth gets hit hardest: “long-duration” cash flows shrink most when the discount rate rises.
- Today’s inputs: 10-yr Treasury 4.40%, implied ERP ~4.23% → ~8.6% for a  $\beta = 1$  stock.

## TERMS ON THIS SLIDE

$r$  — required return  
= risk-free rate + ERP

**ERP** — Equity Risk Premium  
extra return demanded over risk-free

**P/E** — Price-to-Earnings

$g$  — growth rate

**M2**

# **Reading the business**

The three statements, free cash flow, GAAP vs. non-GAAP, and quality-of-earnings red flags.



# The three statements

## Income statement

accrual profit, by layer

- Revenue → gross → operating → net
- Gross = pricing power; operating = efficiency
- Net = after capital structure & taxes

## Balance sheet

a point-in-time snapshot

- Assets = Liabilities + Equity
- Cash, total debt, working capital
- Watch receivables & inventory

## Cash flow

cash, not accruals

- Operating – investing – financing
- OCF adds back D&A and SBC
- Hardest to fake — cash is cash

FCF = operating cash flow – capex — the owner's cash that funds dividends, buybacks, and reinvestment.

# GAAP vs. non-GAAP

Non-GAAP excludes items management deems non-core — most controversially, stock comp.

- Common add-backs: SBC, amortization of acquired intangibles, restructuring, mark-to-market.
- SEC Regulation G (2003): any non-GAAP number must show the comparable GAAP measure and a reconciliation.
- Skeptic's view: SBC is a real, recurring cost — it dilutes shareholders.
- The tell: NVIDIA stopped adding back SBC this quarter (\$1.93B).
  - Effectively conceding stock comp is an operating cost — the standard is tightening.

## TERMS ON THIS SLIDE

**GAAP** — Generally Accepted Accounting Principles  
the standardized U.S. rulebook

**non-GAAP** — a company's own "adjusted" numbers

**SBC** — Stock-Based Compensation

**D&A** — Depreciation & Amortization

## Reading the filing, line by line

|                        |                                                       |                      |
|------------------------|-------------------------------------------------------|----------------------|
| Revenue                | income statement                                      | <b>\$81.6B</b>       |
| Gross margin           | 74.9% × revenue                                       | <b>\$61.1B</b>       |
| Operating / net margin | $\$53.5\text{B} \cdot \$58.3\text{B} \div \text{rev}$ | <b>65.6% · 71.5%</b> |
| Free cash flow         | OCF \$50.3B – capex \$1.8B                            | <b>\$48.6B</b>       |

### The twist — EPS

GAAP \$2.39 &gt; non-GAAP \$1.87

**backwards**

GAAP is flattered by a **\$15.9B non-cash gain** on equity stakes that non-GAAP removes — which is also why net income (\$58.3B) exceeds operating cash flow (\$50.3B). Cash flow is the truth-teller.

### TERMS ON THIS SLIDE

**OCF** — Operating Cash Flow  
cash the core business throws off

**FCF** — Free Cash Flow  
= OCF – capex

**DSO** — Days Sales Outstanding  
= (receivables ÷ rev) × days



# The red-flag check on NVDA

**45 days** ▼ clean

DSO — flat vs. 46 a year ago, so collections are healthy (no channel-stuffing signal)

**+128%** ▲ watch

inventory vs. revenue +85% — the yellow flag; most plausibly a Blackwell/Rubin pre-build

**71.5%**

net margin — but net income > operating cash flow, because a \$15.9B gain is non-cash

M3

# Valuation I — multiples

What each multiple is for, enterprise value, the comp table, and the two traps.

# Which multiple fits which business

- Unprofitable / early growth → P/S or EV/Sales (leverage-neutral).
- Capital-intensive, levered or cyclical → EV/EBITDA or EV/EBIT.
- Banks & asset-heavy → P/B + ROE. Mature / income → P/E + dividend yield.
- Cross-sector P/E comparisons mislead — P/E embeds growth, ROIC, risk and leverage.
  - Read every multiple two ways: vs. the company's own history, and vs. peers.

## TERMS ON THIS SLIDE

**P/E** — Price-to-Earnings

**P/S** — Price-to-Sales  
= market cap ÷ revenue

**EV/EBITDA** — EBITDA = op. income + D&A

**PEG** — P/E ÷ growth %



# EV is the takeover price

EV = market cap + total debt – cash

**NVDA**

net cash \$40B

EV \$4.62T < cap \$4.66T

**AAPL**

net cash \$62B

EV \$4.11T < cap \$4.17T

**KO**

net debt \$31B

EV \$386B > cap \$356B

**F**

net debt \$138B

EV \$194B = **3.4×** cap \$56B

## TERMS ON THIS SLIDE

**EV** — Enterprise Value  
= market cap + debt – cash

P/E and P/S ignore the balance sheet; EV multiples put levered and unlevered firms on the same footing.

# Four names, one screen

| Metric           | NVDA     | AAPL     | KO      | F       | S&P  |
|------------------|----------|----------|---------|---------|------|
| Price            | \$192.53 | \$283.78 | \$82.63 | \$14.13 | —    |
| Market cap       | \$4.66T  | \$4.17T  | \$356B  | \$56B   | —    |
| Enterprise value | \$4.62T  | \$4.11T  | \$386B  | \$194B  | —    |
| Trailing P/E     | 29.5     | 34.4     | 26.0    | N/M     | 27.3 |
| Forward P/E      | 19.4     | 31.2     | 25.0    | 9.6     | 20.1 |
| P/S              | 18.4     | 9.2      | 7.2     | 0.30    | —    |
| Dividend yield   | 0.52%    | 0.37%    | 2.57%   | 4.25%   | —    |
| PEG              | 0.43     | 2.71     | 3.19    | 0.46    | —    |

As of late Jun 2026 · re-pull morning-of · Ford's trailing P/E is N/M — FY2025 net loss.

TERMS ON THIS SLIDE

**TTM** — Trailing Twelve Months

**PEG** — P/E to Growth  
= P/E ÷ growth %

**N/M** — Not Meaningful  
earnings negative or distorted

# Re-rating, and the two traps

- Value trap — a low P/E isn't always cheap (Ford).
  - Trailing P/E is N/M — TTM earnings are negative after a ~\$19.5B special-items charge.
- A high P/E isn't always expensive (NVDA).
  - Forward P/E ~19× — below the S&P's 20× — because earnings are still ~doubling.
- PEG bridges them (Ford 0.46, NVDA 0.43) — but ignores risk and durability.
- The multiple is the question, not the answer: it tells you what the market expects — you judge if that's right.

## TERMS ON THIS SLIDE

**P/E** — Price-to-Earnings

**PEG** —  $P/E \div \text{growth \%}$

**N/M** — Not Meaningful  
earnings negative or distorted

**TTM** — Trailing Twelve Months

**M4**

# Valuation II — DCF

The DCF logic, the WACC discount rate, terminal value, and the reverse-DCF sanity check.

# The DCF logic

A company is worth the present value of its future free cash flows.

- Forecast FCF ~5–10 yrs → pick a discount rate (WACC) → compute a terminal value for the rest.
- Discount each year + TV back to today → sum = enterprise value.
- + net cash / – debt → equity value ÷ diluted shares = intrinsic value per share.
- Honest limits: garbage-in/out, terminal-value dominance, hyper-sensitivity to WACC & g.
  - Triangulate with multiples and a reverse-DCF — never a single point estimate.

## TERMS ON THIS SLIDE

**DCF** — Discounted Cash Flow

**FCF** — Free Cash Flow  
= OCF – capex

**WACC** — Weighted-Avg Cost of Capital

**TV** — Terminal Value  
=  $FCF \times (1+g) \div (WACC - g)$

# Build the rate, then the tail

COST OF EQUITY — CAPM

$$= 4.40\% + 2.20 \times 4.23\%$$

**13.71%** ≈ WACC — NVDA is ~net cash, so no debt weight to blend

Terminal value — worth today ≈ \$2.8T

Forecast years

62% terminal value

38%

Most of the worth sits in a year you can't forecast. Above ~85% is a flag to revisit assumptions.

## TERMS ON THIS SLIDE

**CAPM** — Capital Asset Pricing Model

$$\text{cost of equity} = r_f + \beta \times \text{ERP}$$

**WACC** — Weighted-Avg Cost of Capital

**$\beta$**  — beta  
sensitivity to the market

**TV** — Terminal Value  
 $= \text{FCF} \times (1+g) \div (\text{WACC} - g)$

# What growth is priced in?

| WACC \ g | 3%    | 4%           | 5%    |
|----------|-------|--------------|-------|
| 12.0%    | \$153 | \$165        | \$181 |
| 13.7%    | \$123 | <b>\$130</b> | \$139 |
| 15.0%    | \$106 | \$111        | \$117 |

Base case (20%/yr forecast)

**\$130** vs. **\$192.53**

price

The price implies ~26%/yr FCF growth for a decade.

Reverse DCF reframes "is it expensive?" into an answerable question: **can it grow FCF ~20%+ for 10 years?**

## TERMS ON THIS SLIDE

**DCF** — Discounted Cash Flow

### Reverse DCF

solve today's price for implied growth

**g** — perpetual growth rate  
must be  $\leq \sim 4\%$  (nominal GDP)

**M5**

# Quality & the buy checklist

Economic moats, the ROIC-vs-WACC math, base rates, and a 10-question pre-buy checklist.



# Economic moats — five sources

Morningstar's five durable competitive edges — and how common each is among wide-moat names.

- Intangibles (brands, patents, licenses) — present in ~72% of wide-moat names.
- Cost advantage 47% · switching costs 37% · network effect 20% · efficient scale 12%.
- Width = durability: wide  $\approx$  20 yrs, narrow  $\approx$  10, none = no durable edge.
- The anchors:
  - AAPL switching costs + brand · NVDA the CUDA ecosystem · KO brand + scale
  - Ford no durable moat.

## TERMS ON THIS SLIDE

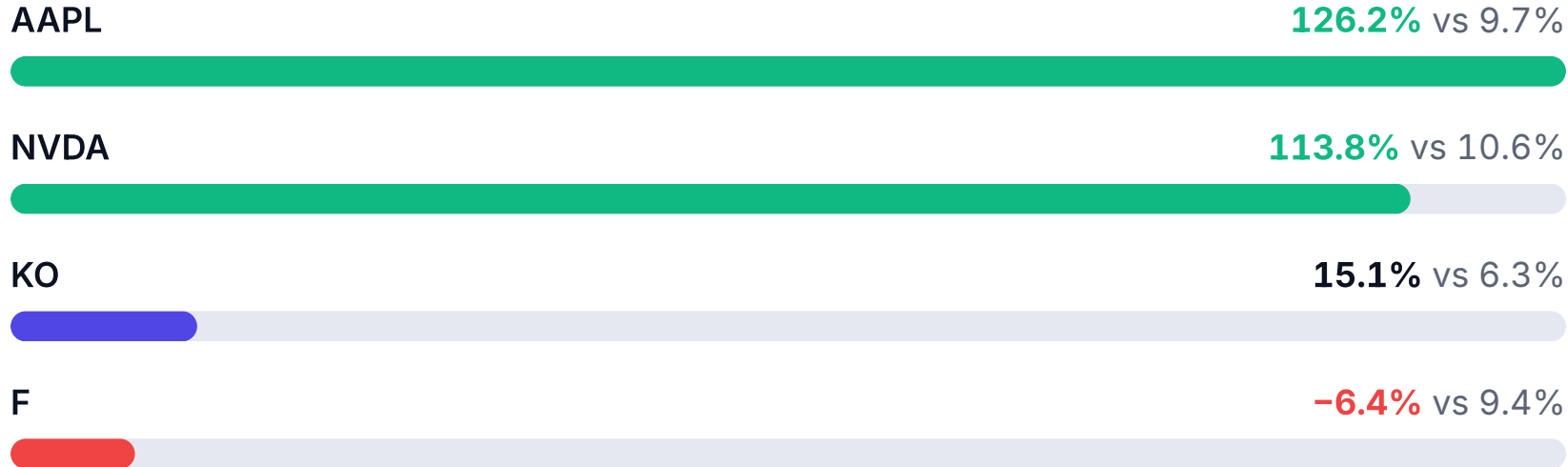
**Moat** — a durable competitive advantage protects excess returns

**TAM** — Total Addressable Market

**CUDA** — NVIDIA's developer software ecosystem

# ROIC must beat WACC

Return on invested capital vs. the cost of that capital — each bar is ROIC; the figure beside it is WACC.



Direction holds under any definition: AAPL, NVDA, KO create value; Ford, at a GAAP loss, destroys it.

## TERMS ON THIS SLIDE

**ROIC** — Return on Invested Capital  
 $= \text{NOPAT} \div \text{invested capital}$

**NOPAT** — Net Operating Profit After Tax

**WACC** — Weighted-Avg Cost of Capital



# Most active pickers trail the index

**~86%** ▼ lag

large-cap funds that trail the S&P 500 over 10 years (91% over 20)

**0%**

top-quartile funds in 2021 that stayed top-quartile through 2025

**79%**

active large-cap funds that lagged in 2025 alone — 4th-worst in 25 years

# The pre-buy checklist

A great business bought at a rich multiple can still be a poor stock — both judgments must clear the bar.

- Moat? Name which of the 5 sources applies — and is it widening or narrowing?
- ROIC > WACC, sustainably? Durable, profitable growth with a long reinvestment runway?
- Margins high & stable? Strong FCF conversion after capex? Healthy balance sheet?
- Disciplined capital allocation — buybacks when cheap, covered dividend, insider alignment?
- Reasonable valuation — what's priced in, and what would break the thesis?
  - Pre-commit the disconfirming signals that would make you sell.

## TERMS ON THIS SLIDE

**ROIC** — Return on Invested Capital

**WACC** — Weighted-Avg Cost of Capital

**FCF** — Free Cash Flow  
= OCF – capex

**M6**

# **How hedge funds & quants play it**

Factors, long/short, event-driven, quant, options and leverage — and whether it beats the index.

# The strategy landscape

Recognize the landscape — not execute it.

- Factor / smart beta: value, momentum, quality, size, low-vol explain most of returns.
- Long/short & market-neutral: buy winners, short losers; size net beta  $\approx 0$ . Shorts have unlimited downside.
- Event-driven (merger arb), global macro, activist (a 13D once ownership tops 5%).
- Quant / stat arb: many tiny statistical edges; alt-data forecasts the print.
  - Options: covered call = income; protective put = insurance. Leverage cuts both ways — see LTCM.

## TERMS ON THIS SLIDE

**13D** — Schedule 13D

filed once ownership tops 5%

**LTCM** — Long-Term Capital Management

1998 leveraged blow-up

**2 and 20** — 2% management + 20% performance fee



# Buffett's \$1M bet, settled

**+125.8%** ▲ winner

a low-cost S&P 500 index fund,  
2008–2017 (~8.5%/yr)

**+2.8–88%** ▼ lost

the five funds-of-funds — low  
single digits annualized

**\$347.51**

GameStop's Jan-2021 squeeze  
close — driven by sentiment, not  
short-covering (SEC)

**M7**

# Technical analysis & timing

What TA claims, the EMH challenge, the honest evidence, and the one robust exception.



# What TA claims — and the challenge

TA is a timing overlay, not a stock-selection system.

- Premises: price & volume already discount the info; trends persist until they reverse.
- The challenge — weak-form EMH (Fama 1970): past prices can't predict future returns.
  - Grossman-Stiglitz: a perfectly efficient market is impossible — so some edge can exist.
- Tools: 50/200-day MAs (golden/death cross), support & resistance, RSI, MACD, volume.
- The burden of proof is on the chartist — markets are “mostly efficient with exploitable frictions.”

## TERMS ON THIS SLIDE

**EMH** — Efficient Market Hypothesis

**RSI** — Relative Strength Index  
0–100 momentum oscillator

**MACD** — Moving Avg Convergence Divergence

**MA** — Moving Average  
50 / 200-day

# The evidence, honestly

- TA overall: mixed-to-weak — most studies compromised by data-snooping and undercounted costs.
- The one robust exception — cross-sectional momentum (Jegadeesh & Titman): ~1.3%/mo.
  - But it mean-reverts and crashes: past losers +163% vs. winners +8% in Mar–May 2009.
- A golden cross is a lagging regime filter — it describes a trend, it doesn't predict one.
- Legitimate use: entry/exit timing on a thesis you already hold — never the reason to own it.

## TERMS ON THIS SLIDE

**MA** — Moving Average  
50 / 200-day

**Golden cross** — 50-day rises above 200-day

**EMH** — Efficient Market Hypothesis

**M8**

# **Building & managing the portfolio**

Goal-based allocation, account types, 2026 tax numbers, and a five-trigger sell discipline.

# Goal → horizon → allocation

- Allocation is driven by when you need the money and how much loss you can absorb — not feelings.
- Short-term (0–3 yr) = preservation: T-bills ~3.7%, money-market ~3.56% (state-tax-exempt).
- Long-term (10+ yr) = equity-heavy; retirement = a glide path that de-risks with age.
- Vanguard's default: ~90% equity early → 30/70 by age 72.
  - "110/120 – age" is a sanity check, not a prescription — age is a crude proxy.

## TERMS ON THIS SLIDE

**T-bill** — Treasury bill  
state-tax-exempt interest

**Glide path** — allocation that de-risks with age

**ERP** — Equity Risk Premium



# Roth vs. traditional

## Traditional

tax-later

- Pre-tax in → deduction now
- Grows tax-deferred
- Taxed as income at withdrawal
- RMDs forced starting in your 70s

## Roth

tax-now

- After-tax in → no deduction
- Qualified withdrawals come out tax-free
- No RMDs, ever
- Backdoor Roth above the income limits

HSA is the only triple tax advantage — pre-tax in, tax-free growth, tax-free out for medical (requires an HDHP).

# Limits & tax rates to know

## CONTRIBUTION LIMITS

|                 |          |
|-----------------|----------|
| 401(k) deferral | \$24,500 |
| IRA / Roth IRA  | \$7,500  |
| HSA family      | \$8,750  |
| Catch-up 50+    | +\$8,000 |

## CAPITAL GAINS

|                    |              |
|--------------------|--------------|
| Long-term (>1 yr)  | 0 / 15 / 20% |
| Short-term (≤1 yr) | ordinary     |
| NIIT surtax        | +3.8%        |
| Loss offset / yr   | \$3,000      |

## TERMS ON THIS SLIDE

**NIIT** — Net Investment Income Tax  
+3.8% above MAGI thresholds

**MAGI** — Modified Adjusted Gross Income

**RMD** — Required Minimum Distribution  
forced from traditional accounts in your 70s

Crossing the 1-year mark can move a \$20k gain from 24% (\$4,800) to 15% (\$3,000) — **\$1,800 saved** on the identical gain.

# When to sell

Five legitimate triggers — most everything else is noise.

- (a) The thesis breaks · (b) valuation / price target hit · (c) a materially better opportunity.
- (d) Rebalancing or position-size management · (e) tax-loss harvesting.
- The core test: would I buy it at today's price? If not, holding is just inertia.
  - This strips out your purchase price entirely.
- Behavioral trap — the disposition effect: sell winners too early, ride losers too long.
  - ~1.5× more likely to sell a winner. But don't let the tax tail wag the dog.

## TERMS ON THIS SLIDE

**Disposition effect** — selling winners early, holding losers

**Wash sale** — loss disallowed if you rebuy within 30 days

**RMD** — Required Minimum Distribution

**Market moment**

# **The most concentrated index in history**

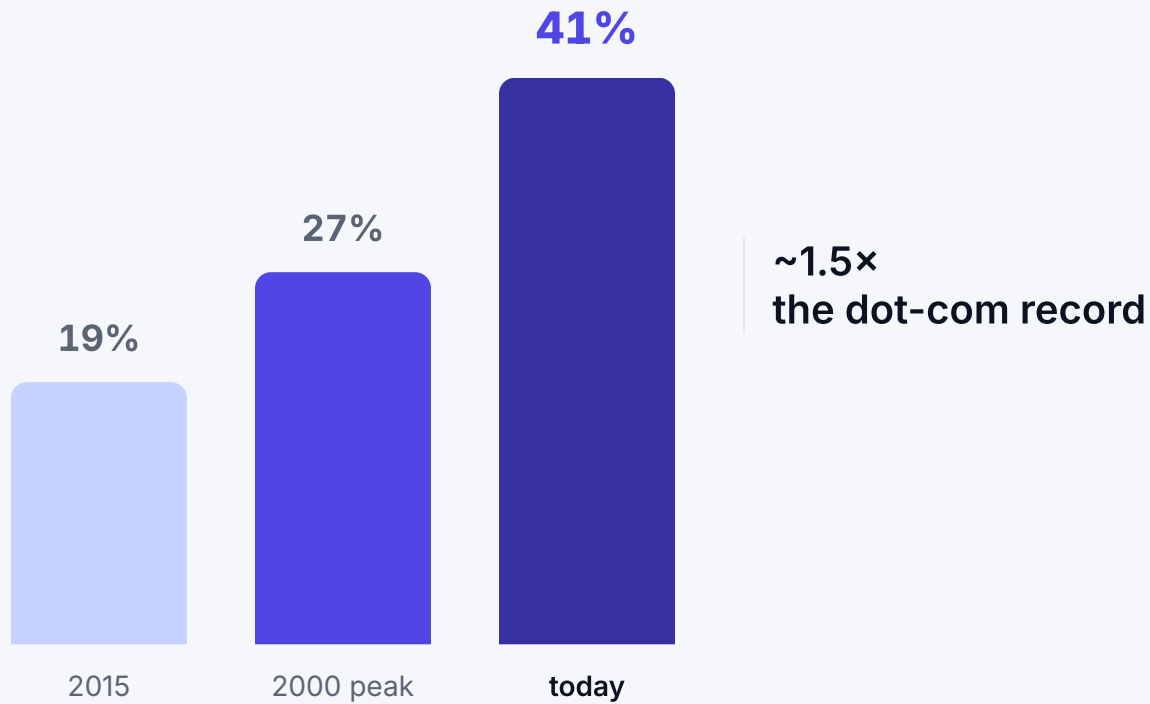
A handful of AI mega-caps now drive the S&P 500's level, valuation, and earnings — and officialdom is flagging it.





# Top 10 = 41% of the S&P 500

Share of index market cap held by the ten largest stocks.



## 32.7%

Magnificent 7 share of the S&P 500 (~\$22T): NVDA 7.4 · Alphabet 6.6 · Apple 6.4 · Microsoft 4.2 · Amazon 3.8 · Tesla 2.2 · Meta 2.1

### TERMS

**Mag 7** — the Magnificent Seven mega-caps



# Cap share is running ahead of earnings share

**41% vs  
32%**

▲  
gap

top-10 share of index weight vs. of earnings — paying up for future growth

**20.1**

S&P 500 forward P/E — above its 5-yr (19.9) and 10-yr (19.0) average

**\$725B** ▲ +77%

2026 hyperscaler capex — increasingly debt-funded

# Why it matters

Index investors are less diversified than they think — and a high multiple is a promise.

- Index investors are less diversified than they think: “the S&P 500” is a ~33% bet on 7 correlated AI names.
- A \$10,000 index buy implicitly holds ~\$3,270 of the Mag 7 — and ~\$740 of NVIDIA alone.
- A high multiple is a promise: NVDA’s PEG implies ~45% growth — beat the whisper, not the consensus.
- Cap share > earnings share is the tell — a multiple-compression year closes that gap.
  - NVDA is down ~18% YTD even as earnings rose 85% — almost pure re-rating in reverse.

## TERMS ON THIS SLIDE

**Mag 7** — the Magnificent Seven mega-caps

**PEG** —  $P/E \div \text{growth } \%$

**YTD** — Year-to-Date

# Discussion

- 1 NVDA beat on every metric and fell — down ~18% YTD while earnings rose 85%. What exactly was “priced in,” and how would you have known the bar was that high before the print?
- 2 The top 10 are 41% of the index's weight but only 32% of its earnings. Justified by superior growth, or concentration risk — and does it change “just buy the S&P 500”?
- 3 Hyperscalers will spend ~\$725B in 2026, increasingly with debt. Using a reverse-DCF mindset, what must that spending produce to justify prices — and what's the contagion path if it doesn't?

# Sources



Compound Money Talk

## WHAT MOVES A PRICE

Mauboussin & Callahan — Total Shareholder Return (Morgan Stanley)

Graham — Mr. Market · Berkshire 1987 letter

Fed H.15 / FRED · Damodaran implied ERP

FactSet Earnings Insight — S&P 500 forward P/E

## STATEMENTS & VALUATION

NVIDIA Q1 FY2027 results & 10-Q

SEC Regulation G — non-GAAP

stockanalysis.com — live multiples

Damodaran — P/E, WACC & ERP datasets

## QUALITY, FUNDS & TECHNICALS

Morningstar — economic moats

SPIVA U.S. Mid-Year 2025 & Persistence

Jegadeesh & Titman — momentum · Fama — EMH

SEC — GameStop report · Fed — LTCM

## PORTFOLIO, TAXES & THE MARKET MOMENT

IRS 2026 limits · HSA & cap-gains Rev. Procs.

Vanguard glide path · Odean — disposition effect

RBC — The Great Narrowing · Mag 7 weights

Fed Financial Stability Report (May 2026) · hyperscaler capex